

Dr. Nico Winkel

Astrophysicist · Data Scientist

Mathematical Optimisation · Energy Systems · Industrial Operations

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PROFILE

I am a data scientist building decision systems for industry and critical infrastructure. At OmegaLambdaTec I lead technical delivery for grid operators and manufacturers, working on mathematical optimisation and energy-system simulation. My work covers the full path from a customer's problem statement to a running system: framing the problem, choosing a method that stays tractable at real scale, and integrating the result into the customer's existing software landscape. My PhD at the Max Planck Institute for Astronomy taught me to own a method end to end and to explain it clearly to experts and non-experts alike.

INDUSTRY EXPERIENCE

07/2025 – now

Data Scientist

Garching, Germany

OmegaLambdaTec GmbH

KI-HEAP4.0 Project Manager & Developer

BayVFP-funded 5-partner consortium with GROB and MEILLER; team of three plus a junior colleague

- Built the CP-SAT model of a flexible job shop with shared transport, pallet and setup-station constraints, including a constraint formulation not previously described for the FJSSP. It reduces makespan by up to 15% and re-plans multi-week horizons within minutes. Its provably optimal schedules serve as the consortium's reference for the partners' reinforcement-learning approach.
- Extended the plant simulation with a stochastic machine-failure module, with alarm rates and repair times derived from historical machine data.
- Designed the integration path from optimiser to shop floor, covering machine event streams, SAP/MES interfaces and a divergence monitor that detects when the executed plan drifts from the optimised one. This decoupled delivery from the client's system-upgrade timeline.
- Owned the domain knowledge transfer from the industrial partners as a continuous process of site visits and working sessions. It produced the use-case portfolio and set what the project built.

Redispatch 2.0 Lead Developer

Customer project for TransnetBW

- Led the implementation of a multi-period MILP that clears transmission-grid congestion by re-dispatching continuous, step-discrete and storage assets under ramp, activation and switching constraints.
- Replaced repeated AC power flow with PTDF-linearised branch flows. This keeps the full horizon tractable across hundreds of plants and 96 timesteps.
- Built a solver-agnostic Pyomo backend with explicit MIP-gap and runtime budgets, reporting congestion relief against redispatch cost per timestep.

RLMate4WP Developer

BMW-funded research project with CAE Würzburg; AI-based heat-pump control

- Implemented RL agents, compared against model predictive control and threshold control for heat pumps with PV and battery, validated on a physical building model.
- In simulation, model predictive control cut annual energy costs by over 30% and energy use by almost 20% against standard control under dynamic tariffs.

SKILLS

Methods	MILP and constraint programming · scheduling (FJSSP) · model predictive control · reinforcement learning (PPO, SAC) · statistical inference · time series analysis · discrete-event simulation · simulation-based evaluation
Modelling & solvers	Pyomo, OR-Tools CP-SAT · HiGHS, SCIP, CBC, GLPK
Engineering	Python (NumPy, Pandas, scikit-learn, SciPy, SimPy), SQL, Bash, Git, Docker, HPC · SAP/MES integration · Matplotlib, Plotly, Dash
Domain	Transmission-grid congestion management (Redispatch 2.0) · heat-pump control and building energy flexibility · industrial production planning
Languages	German (native) · English (C1) · Spanish (A2, learning) · French (A1)

MENTORING & TEACHING

09/2025 – 07/2026	Technical Mentor & Supervisor, for Intern Developer OmegaLambdaTec GmbH	Garching, Germany
10/2020 – 09/2021	Tutorial Lecturer Physics & Astronomy University of Heidelberg	Heidelberg, Germany
02/2018 – 11/2019	Private Tutor Mathematics Studentenring Heidelberg	Heidelberg, Germany
04/2017 – 09/2019	Teaching Assistant Physics University of Heidelberg	Heidelberg, Germany

RESEARCH EXPERIENCE

02/2023 – 04/2026	Referee for Scientific Journals Astronomy & Astrophysics · MNRAS · The Astrophysical Journal	
11/2024 – 02/2025	Postdoctoral Researcher, Astrophysics Max Planck Institute for Astronomy Principal Investigator of a 212 h ($\simeq 1.2$ M€) observing programme with ESO VLT: proposal, strategy and acquisition.	Heidelberg, Germany
10/2020 – 10/2024	Doctoral Researcher, Astrophysics Max Planck Institute for Astronomy - Led data reduction and modelling of TB-scale <i>Hubble</i>, <i>ALMA</i> and <i>JWST</i> datasets; built calibration and inference pipelines in Python for instruments in active use. - Coordinated international collaborations resulting in 16 peer-reviewed publications.	Heidelberg, Germany

EDUCATION

10/2020 – 10/2024	Dr. rer. nat. Astronomy University of Heidelberg · magna cum laude	Heidelberg, Germany
10/2018 – 09/2020	Master of Science in Physics University of Heidelberg · Grade 1.1	Heidelberg, Germany
10/2014 – 09/2018	Bachelor of Science in Physics University of Heidelberg · Grade 2.2	Heidelberg, Germany